COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis
Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

Machine-computed from 16 years of daily price data through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. ***These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. This cycle's overall qualitative score set (2026-09-12_draft) is a light daily catch-up, COMMITTED — AUTO-STAMPED SEPTEMBER 12, 2026: one qualitative pillar re-rated down — Iran War Risk (on Houthi forces sweeping the rest of Yemen's Red Sea coastline and seizing Bab al-Mandeb's Mayyun Island, plus a UKMTO-confirmed strike on two more vessels near Khasab, Oman) — while CLARITY Act, Tokenization RWA, and Fed & Macro held flat — see those pillar reports. This quant pillar's own output is unaffected by qualitative status and enters the master blend the same way regardless.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through September 12, 2026 (BTC $77,289.66) · Pipeline re-run & verified September 12, 2026 (light daily catch-up, 2026-09-12_draft) · Not Financial Advice

The Scores — September 12 Model Output

Computed September 12, 2026 (previous computed run September 11, 2026; GitHub Actions data fetch advanced the cycle-month clock and moved the mapping) — same standard as every qualitative pillar. This cycle's mapping eases at the front end as the 1-month cycle analog fell back: 1mo 58→54 (down, the 1-month cycle analog eased back to 1-of-3 positive, median −0.7%), 3mo 47→47 (flat), 6mo 59→59 (flat), 1yr 68→68 (flat), 3yr 69→69 (flat) — no new weekly, monthly, or quarterly bar closed this cycle (the current week closes Sept 13), so completed-bar composites are unchanged from Sept 11. This cycle's overall qualitative score set (2026-09-12_draft) is a light daily catch-up, COMMITTED — AUTO-STAMPED SEPTEMBER 12, 2026 (Iran War Risk re-rated down moderately as Houthi forces swept the rest of Yemen's Red Sea coastline and seized Bab al-Mandeb's Mayyun Island; CLARITY Act, Tokenization RWA, and Fed & Macro held flat); this quant pillar is computed fresh every run regardless and carries no draft/committed status of its own.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	**54**	0.75	8	**+0.48**	Neutral — the weekly close above the 200-week MA remains confirmed on the completed Sept 6 bar ($80,338.98); the still-forming week (closing Sept 13) ticked up with price; the 1mo cycle-analog eased back to 1-of-3 positive, median −0.7%
3 months	**47**	0.70	8	**−0.34**	Neutral — signal holds negative, still the softest window; analog holds at 1-of-3 positive, median −7.6%
6 months	**59**	0.55	8	**+0.79**	Mild Bull — unanimous 3-of-3 positive, median +38.5%
1 year	**68**	0.60	8	**+1.73**	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3), median +78.2%
3 years	**69**	0.50	8	**+1.52**	Bull — value + cycle (36-month analog 3-of-3 positive, median +1295.7%)

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P − 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

P — the odds the tape is *good for Bitcoin* by then. Near-term still confirmed on a completed-bar basis, and the still-forming week ticked up with price to $77,289.66 (weekly composite as-is +33.0/100, Slight bullish, agreement 5-of-7; completed-bar weekly composite holds at +40.7, unchanged since the weekly bar dated 2026-09-06 closed above both the 20-week and 200-week MAs); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, median +78.2%, and price sits ~45% below its power-law fair value).

Quality — how trustworthy the evidence is. Highest near-term (843 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).

Impact — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: −50% from the all-time high just weeks ago, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price ticked up to $77,289.66 on Sept 12, up from Sept 11's $77,063.25 — +32.0% off the June 30 low; the tape moving on its own schedule is itself information: this cycle's uptick comes even as the Iran pillar's own evidence moved further negative, not from anything in the TA model itself.)

The model: the short-term tape remains genuinely confirmed on a completed-bar basis, and the still-forming week ticked up with price — the weekly composite (as-is) holds Slight bullish (+33.0, agreement 5-of-7) and the quarterly composite holds Slight bullish (+24.6, agreement 4-of-7); the monthly composite ticked up (+1.2, Neutral) with agreement still 1-of-7. No new weekly, monthly, or quarterly bar closed this cycle (the current week closes Sept 13), so completed-bar composites are unchanged: weekly +40.7, monthly +1.8, quarterly +7.7. Repair Signal #1 stays confirmed on the Sept 6 completed bar, unchanged this cycle. The 1-month window eased to Neutral (P 54, down from 58) as the 1-month cycle analog fell back from 2-of-3 to 1-of-3 positive (median −0.7%, from +2.4%) — the same alignment sensitivity this report keeps flagging, in the other direction again; 3 months holds Neutral (P 47, flat) with its analog still 1-of-3 positive (median −7.6%, from −8.0%). The 6-month window holds Mild Bull (P 59) with the analog still unanimous (3-of-3 positive, median +38.5%). The long read holds its direction (1yr flat at 68, 3yr flat at 69): every prior cycle at this exact month was higher 12 months later (median +78.2%), price is ~45% below fair value ($140,025), and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the tape) — and the tape's completed-bar read has stayed "confirmed bullish" throughout this cycle's daily price wobbles. The model still reads three — tape (confirmed at the weekly level, unchanged this cycle), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.78, past the average-bottom date), and valuation (~45% below the power-law fair line, with price sitting well above the −1σ "cheap" band). All three clocks point the same direction on the medium-to-long horizon — the near-term clock is the one doing the day-to-day flexing, exactly as its lower Quality input says it should.

The Facts — The Tape

*Computed September 12, 2026 from daily data (BTC $77,289.66, daily close as of September 12). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite holds at **+40.7** (no new weekly bar closed this cycle; the current week closes Sept 13); monthly +1.8 and quarterly +7.7 are unchanged — no new monthly or quarterly bar closed this cycle.*

- **Weekly composite +33.0/100 (Slight bullish), agreement 5-of-7 indicators** — ticked up as the still-forming week tracked an up day. Price is above the 20-week ($70,102, +10.3%) and well above the **200-week average ($65,174, +18.6%)**; price sits below the 50-week ($79,647, −3.0%). RSI 55.2; MACD line negative (−806) with histogram positive (+2,453) — momentum repair continuing; %B 0.74. Price ($77,289.66) remains above the falling weekly resistance trendline ($75,475) that it broke above earlier this cycle.

★ **Repair signal #1 (a weekly close above the 200-week MA) remains CONFIRMED, unchanged this cycle.** The weekly bar dated 2026-09-06 still closed at $80,338.98, +23.3% above its 200-week average — no new weekly bar has closed since (the current week closes Sept 13). The completed-bar weekly composite holds at **+40.7** — the clearest single confirmation this pillar has logged this cycle.

- **Monthly composite +1.2/100 (Neutral), agreement 1-of-7** — ticked up as September's still-forming bar tracks the uptick. Below the 20-month average ($86,965, −11.1%), above the 50-month ($62,131, +24.4%); RSI 50.4; monthly MACD line +219 with histogram −3,807 — still negative; %B 0.36. Completed-bar monthly composite (August's closed bar) holds at +1.8, unchanged.
- **Quarterly composite +24.6/100 (Slight bullish), agreement 4-of-7** — essentially flat with the cycle-month clock. Comfortably above the 20-quarter average ($58,087, +33.1%) and the 50-quarter ($28,711, +169.2%); RSI 57.0, MACD line +16,128 with histogram −512. Completed-bar quarterly composite is +7.7, unchanged (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price ($77,289.66) sits between the old $74,434 support (−3.7%, 1 touch) and overhead resistance at $81,685 (+5.7%, 2x touch) — the nearest test on the board — then $89,164 (+15.4%, 1x) and

$98,090 (+26.9%, 2x). Further below, $59,930 (−22.5%, 1x) and $57,735 (−25.3%, 1x). The falling weekly support trendline sits at $42,779; the falling resistance trendline, at $75,475, remains broken to the upside, price now holding comfortably above it.

★ **June 30 remains the cycle low: $58,526 close** — the flush tagged the Fib golden pocket ($54,257–$57,802 = 61.8–65% retrace of the $15,479 → $126,272 run) and has since built a **+32.0% bounce to $77,289.66**. **$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.

Weekly tape, scored +33.0/100 (Slight bullish) with agreement 5 of 7 indicators: the weekly close above the 200-week MA remains confirmed on the Sept 6 completed bar, price holds comfortably above the falling resistance trendline, MACD histogram remains positive. Machine-annotated from the same data that feeds the scores.

The Facts — The Cycle Clock & The Value Floor

The two slower clocks. Same data, longer lenses.

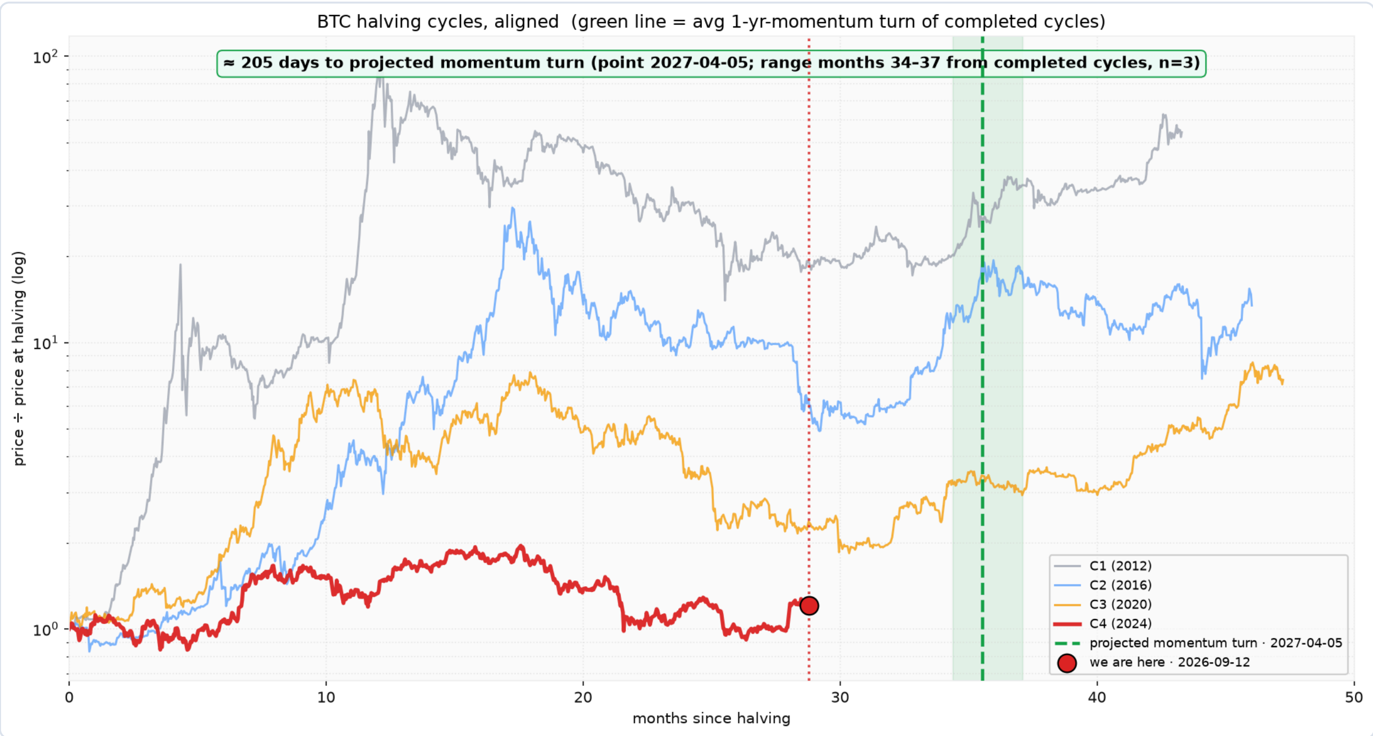
- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — $124,728) and printed its low-so-far at month 26 (June 30, $58,526, −53% from the top). **We are at month 28.78 — past the average-bottom date.**

★ **The cycle analogs are the spine of the long-window scores.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **1-of-3** (median −0.7%) · next 3 months positive **1-of-3** (median −7.6%) · next 6 months positive **3-of-3** (median +38.5%) · next 12 months positive **3-of-3** (median +78.2%) · next 36 months positive **3-of-3** (median +1295.7%). *Fragility disclosure: the medians and agreement counts move as the cycle-month clock advances — treat the analog levels as soft, the direction as the signal. This cycle's 1-month analog fell straight back from 2-of-3 to 1-of-3 positive on a single day's clock advance — after flipping the other way the day before —*

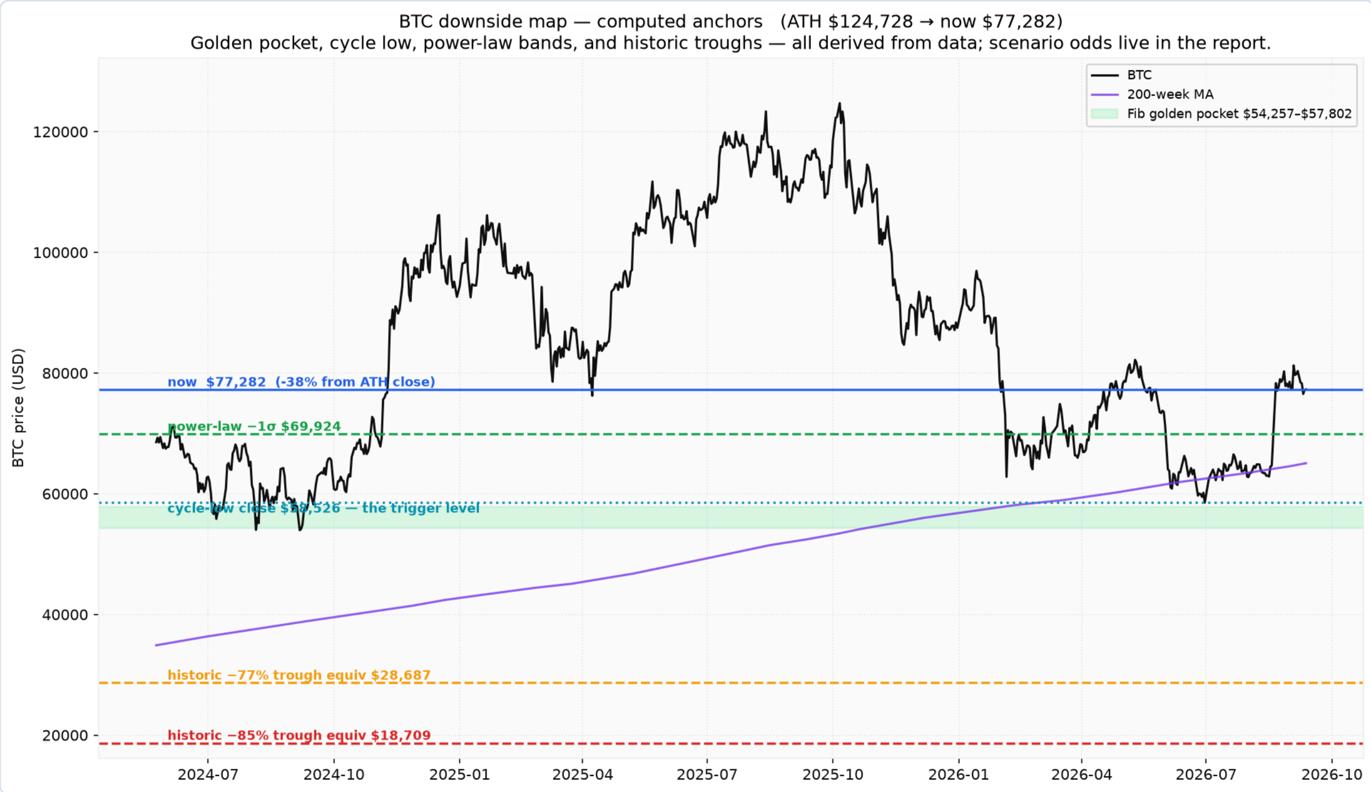
exactly the kind of alignment sensitivity this disclosure exists to flag. And know what the 36-month number is: from month ~28 that window reaches month ~64 — it contains the NEXT halving's rally.

- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5)** → still projects to **~April 2027** (80% window Jan–Jul 2027) for this cycle.
- **The power-law floor:** a log-log fit over the full price series puts fair value at **$140,025** — price is **−45%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value materially (fair value ranges roughly $118K–$145K depending on fit window) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price ($77,289.66) sits above the **−1σ band ($69,924)** — out of the "cheap" zone (the −2σ band sits at $34,918, far below). Every halving-era cycle trough was −77% to −85% from ATH ($18.7K–$28.7K equivalent today) — the pre-halving 2011 bear went ~−93% (~$8.7K equivalent), excluded from the range but in the data; this cycle's low-so-far was −53% from the ATH.
- **No froth in the system:** the altcoin-froth index reads z = −0.87 — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to −39% average 12-month drawdowns vs −23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23, unchanged) — used as a tilt, not a score input.*

★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 28.78, past the historical bottoming window.



The downside map, updated after the June 30 flush: the projected $54K–$61K bottom zone was hit ($58,526 close / $57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to $54–58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From $77,289.66: the 20-week average (~$70.1K) is well behind us — price sits **+10.3%** above it; the 50-week average ($79,647) is back overhead as the immediate test, now **3.0%** away — while the falling weekly trendline resistance ($75,475) remains broken to the upside, held comfortably. A retest of the $124,728 ATH is **+61.4%**; the power-law fair line is **+81.2%**. The median 12-month analog from this cycle-month is now **+78.2%** (down from Sept 11's +79.0% read — a reminder the analog levels move as the alignment shifts); the median 36-month analog is +1295.7% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the near-term signal remains confirmed on a completed-bar basis, though it eased this cycle as the 1-month cycle analog fell back — the near quarter is still the softest window and holds a negative signal (P 47), but the 6-month-and-beyond case remains unanimous (3-of-3) and holds steady (6mo P 59).

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A −53% drawdown at month 26 vs −77/−85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is $18.7K–$28.7K, far below every level in this report.

- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket $54.3–57.8K is the target; re-rate 1/3-month down
~$65,173	Weekly close above the 200-week MA	**CONFIRMED** — the weekly bar dated 2026-09-06 closed at $80,338.98, +23.3% above this level; completed-bar weekly composite holds at +40.7, unchanged this cycle (no new weekly bar yet)
~$79,647	Weekly close above the 50-week MA	Price sits −3.0% below it — the next real test, a touch closer after an up day; a confirmed weekly close above it would extend trend repair further — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) has now passed (we're at month 28.78)	Price sits well off the June 30 low (+32.0%); a higher low + improving monthly momentum after this date = basing confirmed — August's completed monthly bar (+1.8) is an early such sign
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
$50,000	Weekly close below	−60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone ($18.7K–$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: COMMITTED — AUTO-STAMPED SEPTEMBER 12, 2026 (2026-09-12_draft2, second cycle of the day) · THIS PILLAR: COMPUTED SEPTEMBER 12, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 54 / 47 / 59 / 68 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read eased this cycle as price ticked up: no new weekly bar closed (the completed-bar weekly composite holds at +40.7, unchanged), and the as-is weekly composite ticked up to +33.0 (5-of-7). The 1-month window eased to Neutral (P 54, down from 58) as the 1-month cycle analog fell back to 1-of-3 positive (median −0.7%) — a one-day clock advance reversing the prior day's flip, which is the n=3 fragility this report keeps disclosing, not a new signal. The 6-month window holds Mild Bull (P 59) with its analog still unanimous (3-of-3 positive, median +38.5%). The 3-month window holds Neutral (P 47, flat) — still the softest window in the set, its analog holding at 1-of-3 positive. The long read holds its direction (1yr flat at 68, 3yr flat at 69) as the cycle-month clock advanced to 28.78 — the direction, not the level, is the signal.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: this cycle's overall qualitative score set (2026-09-12_draft) is a light daily catch-up, COMMITTED — AUTO-STAMPED SEPTEMBER 12, 2026 (Iran War Risk re-rated down moderately as Houthi forces swept the rest of Yemen's Red Sea coastline and seized Bab al-Mandeb's Mayyun Island; CLARITY Act, Tokenization RWA, and Fed & Macro held flat). This pillar's own computed output is unaffected by that process and enters the master blend the same way it does every cycle.

What Changed Since Last Check

Pipeline re-run September 12, 2026 (previous computed run September 11, 2026).

- **Today's refresh in one line:** the GitHub Actions data fetch moved BTC from $77,063.25 (Sept 11) to $77,289.66 (Sept 12), cycle-month advanced 28.75→28.78, no new weekly/monthly/quarterly bar closed (the completed-bar weekly composite holds at +40.7; monthly +1.8 and quarterly +7.7 unchanged), and the mapping eased at 1mo P 58→54 (the 1-month cycle analog fell back negative) while 3mo/6mo/1yr/3yr held flat at 47/59/68/69 — normal daily drift, not a new signal.
- **Price: $79,052.40 (Sept 9) → $77,993.99 (Sept 10) → $77,063.25 (Sept 11) → $77,289.66 (Sept 12).** This cycle's modest uptick comes even as the Iran pillar's own evidence moved further negative (Houthi forces seizing Bab al-Mandeb's Mayyun Island), not a new TA event. June 30's $58,526 remains the cycle low; price is now +32.0% off it.
- ★ **Repair signal #1 remains CONFIRMED, unchanged this cycle: the weekly bar dated 2026-09-06 closed at $80,338.98, +23.3% above the 200-week average.** No new weekly bar has closed since (the current week closes Sept 13) — the completed-bar weekly composite holds at +40.7.
- **No new weekly, monthly, or quarterly bar closed this cycle** — completed-bar composites are unchanged: weekly +40.7 (confirmed since the Sept 6 weekly close above the 200-week MA), monthly +1.8 (August's closed bar), quarterly +7.7 (current bar still forming). On the current (forming) bars, versus Sept 11: weekly +32.5→+33.0 (agreement holds at 5-of-7, still Slight bullish), monthly +1.0→+1.2 (agreement holds at 1-of-7, still essentially flat), quarterly +24.4→+24.6 (agreement holds at 4-of-7, still Slight bullish) — all three ticking up as the still-forming bars track the uptick.
- **Model P vs Sept 11: 58/47/59/68/69 → 54/47/59/68/69.** 1mo −4 on the analog falling back negative; 3mo, 6mo, 1yr and 3yr flat.

- **Cycle analogs moved this cycle as the clock advanced to 28.78:** the 1-month analog's agreement count fell back from 2-of-3 to 1-of-3 positive (median +2.4%→−0.7%); 3-month held at 1-of-3 positive with the median firming slightly (−8.0%→−7.6%); 6/12/36-month held unanimous (3-of-3) with medians moving (6mo +41.7%→+38.5%, 1yr +79.0%→+78.2%, 3yr +1276.1%→+1295.7%) — flagged per this report's standing n=3 fragility disclosure, not treated as new structural information.
- **Power-law fair value: $139,939 (Sept 11) → $140,025 (Sept 12)** (price still −45% vs. fair) — normal drift as new daily bars extend the fit.
- **Data refreshed:** daily price data through September 12, 2026 (data/btc_usd_daily.csv, via the GitHub Actions fetch workflow; this sandbox's egress block prevents running `run.py quant` directly, per the documented workaround); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).
- **Context:** this cycle's overall qualitative score set (2026-09-12_draft) is a light daily catch-up, COMMITTED — AUTO-STAMPED SEPTEMBER 12, 2026 — Iran War Risk re-rated down moderately as Houthi forces swept the rest of Yemen's Red Sea coastline and seized Bab al-Mandeb's Mayyun Island; CLARITY Act, Tokenization RWA, and Fed & Macro held flat. This pillar's own computed output is independent of that process and enters the master blend the same way regardless of the qualitative set's status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via fetch_btc_history.py · Coin Metrics community reference price (2010–2011 close-only) via build_master_history.py — 5,838 days total, each row tagged by source.
- **Indicator stack:** indicators.py — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. levels.py — swing-cluster S/R + least-squares trendlines.
- **Scoring:** score.py — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** cycle_analysis.py — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · downside.py / downside_map.py — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · test_altcoin_froth.py — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** pillar_mapping.py — window scores from tape composites + cycle analogs + power-law value; P = 50 + 25×(S/100); Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Sept 12 (completed-bar composites: weekly +40.7, confirmed by the Sept 6 weekly close above the 200-week MA / monthly +1.8, August's bar having closed / quarterly +7.7, current bar still forming — all three unchanged this cycle since no new bar closed) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.